**Unit:** Unit 6: Systems of Linear Equations and Inequalities **Name:** ________________
**Topic:** Applications of Systems of Inequalities **Date:** ____________

# Advanced (Habanero)

**Q1.** What is the system of inequalities that models the following constraints: $x \geq 0$, $y \geq 0$, and $x + y \leq 10$?
(A) $x \geq 0, y \geq 0, x + y \leq 10$
(B) $x \leq 0, y \leq 0, x + y \geq 10$
(C) $x \geq 0, y \leq 0, x + y \leq 10$
(D) $x \leq 0, y \geq 0, x + y \geq 10$
(E) $x \geq 0, y \geq 0, x + y \geq 10$

**Q2.** A company produces two products, A and B. The profit on product A is

$$3$$

and on product B is

$$5$$

. The company has 120 hours of labor and 80 hours of machinery available. If $x$ is the number of units of A produced and $y$ is the number of units of B produced, which system of inequalities models the situation?
(A) $x \geq 0, y \geq 0, 3x + 2y \leq 120, 2x + 3y \leq 80$
(B) $x \geq 0, y \geq 0, 2x + 3y \leq 120, 3x + 2y \leq 80$
(C) $x \geq 0, y \geq 0, 2x + 3y \leq 120, 3x + 2y \geq 80$
(D) $x \geq 0, y \geq 0, 3x + 2y \leq 120, 2x + 3y \geq 80$
(E) $x \geq 0, y \geq 0, 3x + 2y \geq 120, 2x + 3y \leq 80$

**Q3.** Which of the following points is a solution to the system of inequalities $x \geq 0$, $y \geq 0$, and $x + 2y \leq 6$?
(A) $(0, 3)$
(B) $(2, 3)$
(C) $(3, 0)$
(D) $(0, 0)$
(E) $(4, 1)$

**Q4.** A farmer has 480 square feet of space to plant wheat and barley. Each wheat plant requires 8 square feet and each barley plant requires 12 square feet. If $x$ is the number of wheat plants and $y$ is the number of barley plants, which inequality models the space constraint?
(A) $8x + 12y \leq 480$
(B) $8x + 12y \geq 480$
(C) $12x + 8y \leq 480$
(D) $12x + 8y \geq 480$
(E) $8x - 12y \leq 480$

**Q5.** A bakery sells two types of bread, whole wheat and white bread. The profit on a loaf of whole wheat is

$$0.50$$

and on a loaf of white bread is

$$0.60$$

. The bakery can produce at most 500 loaves per day. If $x$ is the number of whole wheat loaves and $y$ is the number of white bread loaves, which system of inequalities models the situation?
(A) $x \geq 0, y \geq 0, x + y \leq 500$
(B) $x \geq 0, y \geq 0, x + y \geq 500$
(C) $x \leq 0, y \leq 0, x + y \leq 500$
(D) $x \leq 0, y \geq 0, x + y \geq 500$
(E) $x \geq 0, y \leq 0, x + y \leq 500$

**Q6.** What is the maximum value of $3x + 2y$ subject to the constraints $x \geq 0$, $y \geq 0$, and $x + 2y \leq 6$?
(A) 12
(B) 10
(C) 9
(D) 8
(E) 6

**Q7.** A city has $12,000$ dollars to spend on two types of security cameras. Type A costs 200 dollars and type B costs 300 dollars. If $x$ is the number of type A cameras and $y$ is the number of type B cameras, which inequality models the budget constraint?
(A) $200x + 300y \leq 12000$
(B) $200x + 300y \geq 12000$
(C) $300x + 200y \leq 12000$
(D) $300x + 200y \geq 12000$
(E) $200x - 300y \leq 12000$

**Q8.** A company produces two products, X and Y. The profit on product X is

$$8$$

and on product Y is

$$10$$

. The company can produce at most 400 units of X and 300 units of Y per day. If $x$ is the number of units of X produced and $y$ is the number of units of Y produced, which system of inequalities models the situation?
(A) $x \geq 0, y \geq 0, x \leq 400, y \leq 300$
(B) $x \geq 0, y \geq 0, x \geq 400, y \geq 300$
(C) $x \leq 0, y \leq 0, x \leq 400, y \leq 300$
(D) $x \leq 0, y \geq 0, x \leq 400, y \geq 300$
(E) $x \geq 0, y \leq 0, x \leq 400, y \leq 300$

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**Open Ended Questions (FRQ)**

**Q9.** A small business has 100 feet of fencing to enclose two rectangular gardens, one for vegetables and one for flowers. Each garden must have an area of at least 50 square feet. If $x$ is the length of the vegetable garden and $y$ is the length of the flower garden, find the system of inequalities that models the situation and determine the maximum possible area for the gardens. Show all work and explain your reasoning.

**Q10.** A manufacturer produces two types of furniture, chairs and tables. The profit on a chair is



$$20$$

and on a table is

$$30$$

. The manufacturer has 200 hours of labor and 150 hours of machinery available per day. If $x$ is the number of chairs produced and $y$ is the number of tables produced, find the system of inequalities that models the situation and determine the maximum possible profit. Show all work and explain your reasoning.

**Chef's Tips**

- Read each question carefully and choose the correct answer.
- Show all work for free response questions.
- Use a ruler to draw graphs and label axes clearly.